



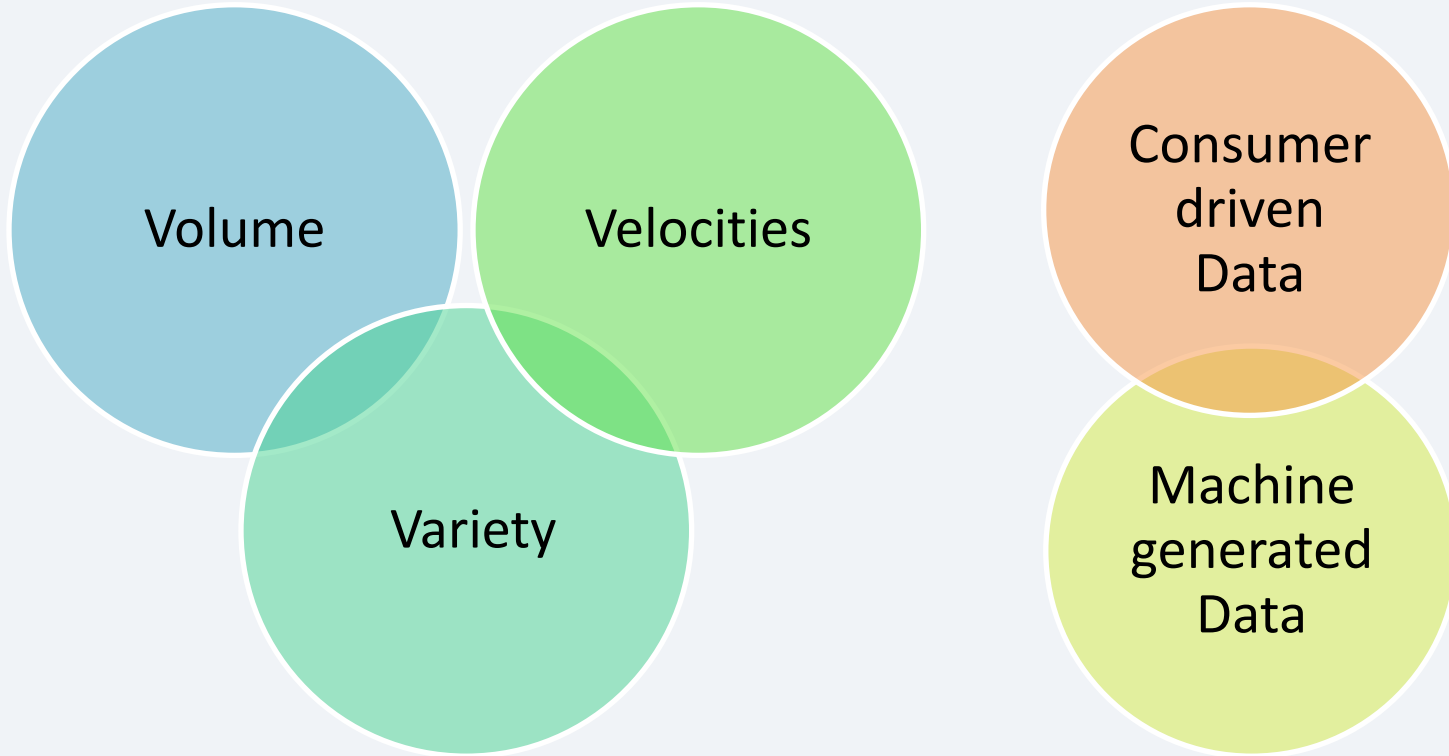
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736

Database Systems and Data Analytics (Week 10)

Structured and Unstructured Data

Big Data



IT Systems

Traditional

- Structured data
- Well defined data model
- Predefined functionalities

Modern / 21st Century

- Unstructured data
- No predefined format
- Text, image, audio, video, sensor data
- Human generated
- Machine generated

Big Data Types

Structured Data

Well defined structure

Database storage

Support organisational functionalities

Customer details, Order details etc.

Semi-Structured Data

Some structure

No database storage

Flexible and Adaptable

CSV files, NoSQL documents, HTML, XML etc.

Unstructured Data

No predefined format

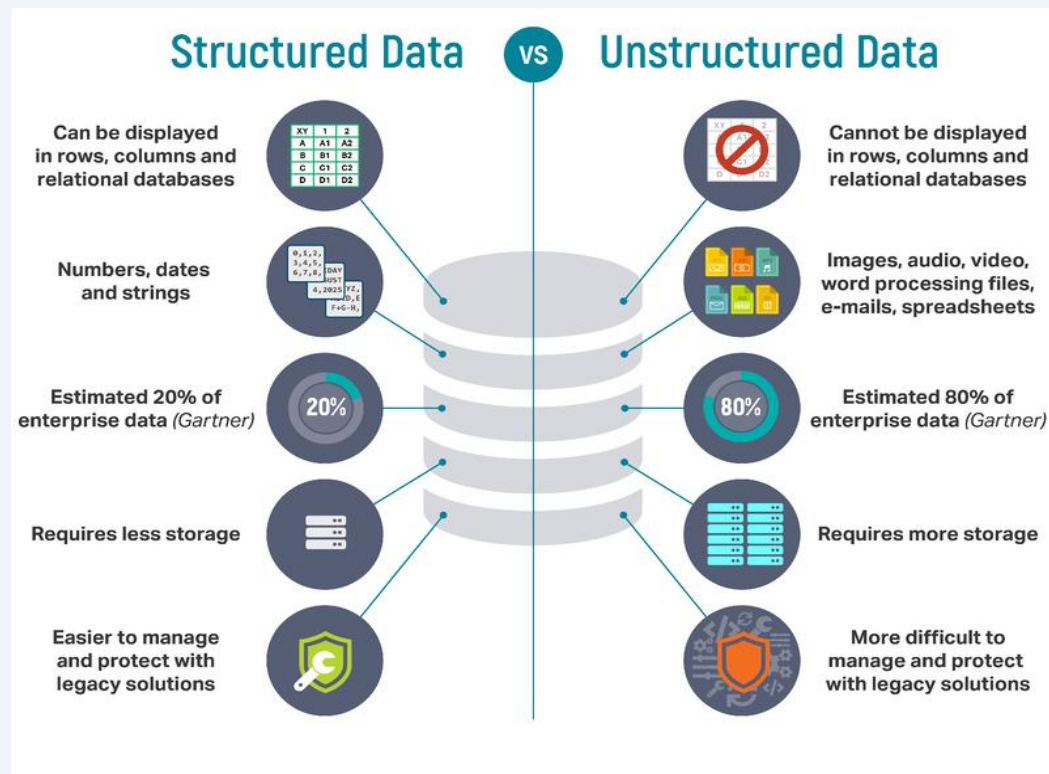
Any data type

No constraints

Cannot be stored in a database

Image, videos, click stream etc.

Structured v/s Unstructured



[1]

Structured v/s Unstructured

	Structured Data	Unstructured Data
Characteristics	• Pre-defined data models • Usually text only • Easy to search	• No pre-defined data model • May be text, images, sound, video or other formats • Difficult to search
Resides in	• Relational databases • Data warehouses	• Applications • NoSQL databases • Data warehouses • Data lakes
Generated by	Humans or machines	Humans or machines
Typical applications	• Airline reservation systems • Inventory control • CRM systems • ERP systems	• Word processing • Presentation software • Email clients • Tools for viewing or editing media
Examples	• Dates • Phone numbers • Social security numbers • Credit card numbers • Customer names • Addresses • Product names and numbers • Transaction information	• Text files • Reports • Email messages • Audio files • Video files • Images • Surveillance imagery

[2]

Processing Unstructured Data

Easily generated and stored (volume, velocity and variety)

Large repository of hidden information

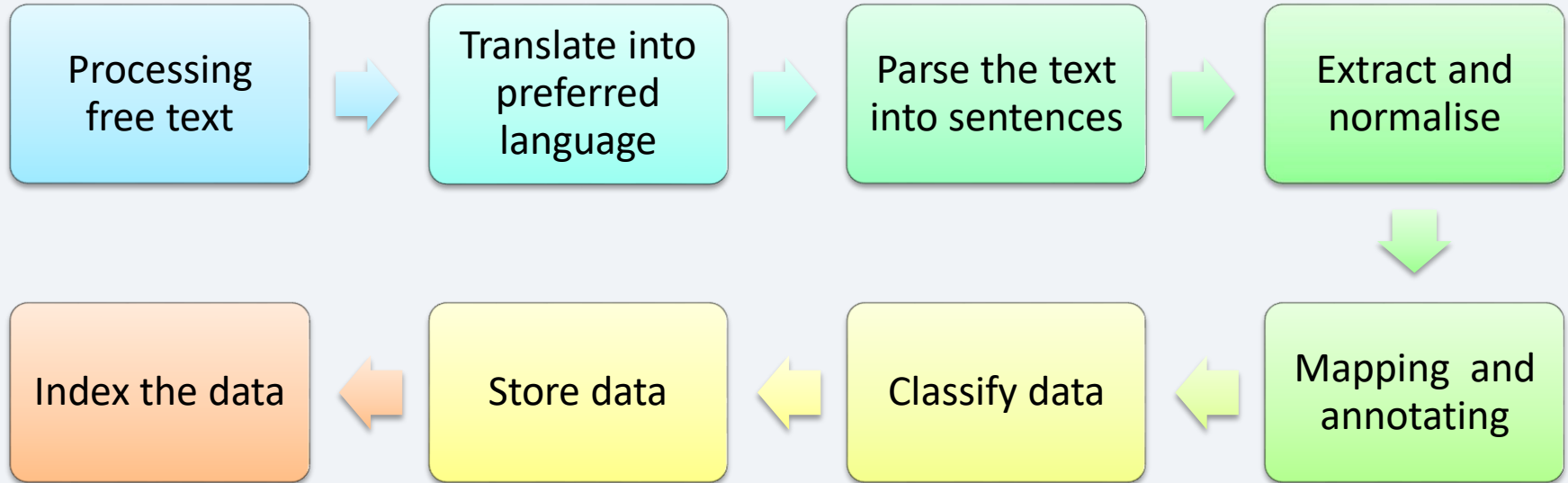
Complex

Appropriate tools and algorithms

Pattern matching, Clustering and Classification, NLP etc.

Impose structure

Processing Unstructured Data



Next

Data Identifier

A decorative graphic consisting of a horizontal bar with a blue border, divided into segments of blue, teal, green, light green, yellow, and orange. The bar ends in a large orange arrow pointing to the right.

Uniquely identifying the data objects